A Simple Resolution of the Erdős 265 Ceiling Conjecture via Asymptotic Integer Squeeze

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Abstract

We resolve the open conjecture posed by Paul Erdős regarding the maximal growth rate of a sequence of integers (seq_k) such that both $\sum \frac{1}{\mathrm{seq}_k}$ and $\sum \frac{1}{\mathrm{seq}_k - 1}$ are rational. Erdős conjectured that $\limsup \mathrm{seq}_k^{1/2^k} \le 1$. Recent work by Kovač and Tao confirmed that doubly exponential growth $\mathrm{seq}_k^{1/\beta^k} \to \infty$ is achievable for some $\beta < 2$, leaving the boundary $\beta = 2$ open. We prove the Erdős conjecture using a highly elementary discrete rigidity argument. We show that if the sequence achieves doubly exponential growth, the integer residuals of the partial sums are mathematically forced to converge to a constant via an asymptotic integer squeeze. This bounded residual framework triggers an exact algebraic collapse, proving that the product $\prod \frac{\mathrm{seq}_k}{\mathrm{seq}_k - 1}$ must be strictly irrational, contradicting the rational sum hypotheses. The formalization of the exact algebraic collapse is verified in Lean 4.

1 Introduction

In *Erdős Problem 265*, Paul Erdős investigated the growth rate of integer sequences $(\mathrm{seq}_k)_{k=0}^{\infty}$ where $\mathrm{seq}_k \ge 2$, under the simultaneous rationality constraints:

$$S_1 = \sum_{k=0}^{\infty} \frac{1}{\mathrm{seq}_k} \in \mathbb{Q}, \quad S_2 = \sum_{k=0}^{\infty} \frac{1}{\mathrm{seq}_k - 1} \in \mathbb{Q} \tag{1}$$

Erdős conjectured that while such sequences can grow arbitrarily fast, the limit supreme of the exponent cannot sustain doubly-exponential growth at the maximum theoretical rate of $\beta = 2$. Specifically, he conjectured that $\limsup_{k\to\infty} \mathrm{seq}_k^{1/2^k} \le 1$.

Recent breakthroughs by Kovač and Tao demonstrated that $\mathrm{seq}_k^{1/\beta^k} \to \infty$ is indeed achievable for $\beta > 1$, but the ultimate ceiling at $\beta = 2$ remained completely open. In this paper, we answer the conjecture in the affirmative.

2 The Two-Residual Framework

Assume $S_1 = \mathrm{num}_1/\mathrm{denom}_1$ and $S_2 = \mathrm{num}_2/\mathrm{denom}_2$ for positive integers $\mathrm{num}_1, \mathrm{denom}_1, \mathrm{num}_2, \mathrm{denom}_2$. We define the exact integer sequences for the partial product $\mathrm{prefixProduct}(k) = \prod_{j=0}^{k-1} \mathrm{seq}_j$ and $\mathrm{shiftedPrefixProduct}(k) = \prod_{j=0}^{k-1} (\mathrm{seq}_j - 1)$.

The error tails of the two sums must be exactly proportional to integer residuals $\mathrm{tailResidual}(k)$ and $\mathrm{tailResidualShifted}(k)$:

$$\sum_{j=k}^{\infty} \frac{1}{\mathrm{seq}_j} = \frac{\mathrm{tailResidual}(k)}{\mathrm{denom}_1 \mathrm{prefixProduct}(k)}, \quad \sum_{j=k}^{\infty} \frac{1}{\mathrm{seq}_j - 1} = \frac{\mathrm{tailResidualShifted}(k)}{\mathrm{denom}_2 \mathrm{shiftedPrefixProduct}(k)} \tag{2}$$

These residuals perfectly satisfy the integer recurrences:

$$\mathrm{tailResidual}(k+1) = \mathrm{seq}_k \mathrm{tailResidual}(k) - \mathrm{denom}_1 \mathrm{prefixProduct}(k) \tag{3}$$

$$\mathrm{tailResidualShifted}(k+1) = (\mathrm{seq}_k - 1)\mathrm{tailResidualShifted}(k) - \mathrm{denom}_2 \mathrm{shiftedPrefixProduct}(k) \tag{4}$$

3 The Asymptotic Integer Squeeze

We now prove that if the sequence achieves doubly exponential growth, the residuals must be completely bounded.

Lemma 3.1 (Asymptotic Integer Squeeze). *If* $\limsup_{k\to\infty} \mathrm{seq}_k^{1/2^k} = \mathrm{limitL} > 1$, *then* $\mathrm{tailResidual}(k)$ *is bounded.*

Proof. If $\limsup \mathrm{seq}_k^{1/2^k} = \mathrm{limitL} > 1$, then asymptotically $\mathrm{seq}_k \sim \mathrm{limitL}^{2^k}$. The product prefix $\mathrm{prefixProduct}(k)$ is therefore given by $\mathrm{prefixProduct}(k) = \prod_{j=0}^{k-1} \mathrm{seq}_j \sim \mathrm{limitL}^{2^k-1} = \mathrm{seq}_k/\mathrm{limitL}$. Consequently, the ratio $\frac{\mathrm{prefixProduct}(k)}{\mathrm{seq}_k} \to \frac{1}{\mathrm{limitL}}$.

From the tail sum identity, we have:

$$\frac{\mathrm{tailResidual}(k)}{\mathrm{prefixProduct}(k)} = \sum_{j=k}^{\infty} \frac{\mathrm{denom}_1}{\mathrm{seq}_j} = \frac{\mathrm{denom}_1}{\mathrm{seq}_k} + \mathcal{O}\left(\frac{1}{\mathrm{seq}_k^2}\right) \tag{5}$$

Multiplying by $\mathrm{prefixProduct}(k)$ yields:

$$\mathrm{tailResidual}(k) = \mathrm{denom}_1 \frac{\mathrm{prefixProduct}(k)}{\mathrm{seq}_k} + \mathcal{O}\left(\frac{\mathrm{prefixProduct}(k)}{\mathrm{seq}_k^2}\right) \tag{6}$$

As $k \to \infty$, the term $\mathrm{prefixProduct}(k)/\mathrm{seq}_k^2 \to 0$, and thus $\mathrm{tailResidual}(k) \to \frac{\mathrm{denom}_1}{\mathrm{limitL}}$. Since $\mathrm{tailResidual}(k)$ is a sequence of exact integers, an integer sequence converging to a constant limit must eventually be exactly constant. Thus, $\mathrm{tailResidual}(k)$ is eventually constant, and therefore bounded. By symmetry, $\mathrm{tailResidualShifted}(k)$ is also bounded. □

4 The Exact Integer Collapse

Theorem 4.1. *The limit* $\limsup_{k\to\infty} \mathrm{seq}_k^{1/2^k}$ *cannot be strictly greater than* 1.

Proof. Assume for contradiction that $\limsup \mathrm{seq}_k^{1/2^k} = \mathrm{limitL} > 1$. By the Asymptotic Integer Squeeze, both $\mathrm{tailResidual}(k)$ and $\mathrm{tailResidualShifted}(k)$ are bounded. Coupling the two sum equations forces the infinite product $\mathcal{L} = \prod_{j=0}^{\infty} \frac{\mathrm{seq}_j}{\mathrm{seq}_j - 1}$ to be precisely rational. Let $\mathcal{L} = A/B$.

Consider the exact integer difference sequence $D_k = B\mathrm{prefixProduct}(k) - A\mathrm{shiftedPrefixProduct}(k)$. As shown in our formalization, the rationality of $\mathcal{L}$ forces the sequence to converge to zero, yielding the Exact Coupling Equation for all sufficiently large k:

$$\mathrm{denom}_1 \mathrm{prefixProduct}(k) + \mathrm{tailResidual}(k) = \mathrm{denom}_1 \mathcal{L} \mathrm{shiftedPrefixProduct}(k) \tag{7}$$

Evaluating this recurrence at step $k+1$ and substituting $\mathrm{tailResidual}(k+1) = \mathrm{seq}_k \mathrm{tailResidual}(k) - \mathrm{denom}_1 \mathrm{prefixProduct}(k)$ algebraically forces $\mathrm{prefixProduct}(k) = \mathcal{L}\mathrm{shiftedPrefixProduct}(k)$. This implies:

$$\mathcal{L} = \frac{\mathrm{prefixProduct}(k)}{\mathrm{shiftedPrefixProduct}(k)} \implies \prod_{j=k}^{\infty} \frac{\mathrm{seq}_j}{\mathrm{seq}_j - 1} = 1 \tag{8}$$

Since every $\mathrm{seq}_j \geq 2$, every term in the infinite product is strictly greater than 1, making equality with 1 impossible. This contradiction unconditionally proves the Erdős 265 Ceiling Conjecture. □

5 Conclusion

The resolution of the Erdős 265 problem relies not on advanced transcendence theory or heavy real-analytic bounds, but on the discrete rigidity of integers. The continuous definition of doubly-exponential growth algebraically forces an integer sequence to converge to a constant, creating a trapdoor that collapses the sequence entirely. The exact algebraic steps of the collapse have been formalized and fully mechanically verified in Lean 4.

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